

Colorful Style

Theorem 1 (*Pythagorean*)

In a right triangle with legs a and b and hypotenuse c :

$$a^2 + b^2 = c^2$$

Definition 1

A **continuous function** is one where small changes in input produce small changes in output.

Lemma 1

If f is continuous at a , then f is bounded in some neighborhood of a .

Corollary 1

Every polynomial is continuous on $\mathbb{R}$.

Remark

This result generalizes to higher dimensions.

Example 1

Consider $f(x) = x^2$. Then $f(2) = 4$.

Proof. By direct calculation and the definition of continuity.

□